Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate concern of Glass up down noise for all door in Marazzo model "

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**10.06.18**	**15.06.2018**
Completion Date	**15.12.18**	**05.12.2018**

Current Status: Closed

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Prasad Deore	L7-0	STCF	9764005447
2	**Mentor**	Chikhale Sandeep	L5-M	STCF	9922408138
3	**Member 1**	Mangesh Musale	L7-0	MQA	9730163917
4	**Member 2**	Amar Pande	L7-O	PRD	9823232181
5	**Member 3**	Dinesh Nagrale	L8-0	STCF	9923327676
6	**Member 4**	Satish Pagar	L8-O	MQA	9822209653
7	**Member 5**	Darshak Thakkar	L7-0	SQA	8866114994

Step1. Select Subject

Brief description of Problem:-

Glass up down noise for all door in Marazzo model.

What : Glass up down noise for all door. .

Where : At TCF CAI & LSP.

When : Since Start of Marazzo model

Who : By associate at regulator stage , Buy off check man & LSP insp.

WHY : This is V2 severity defect with High rework, FRC /FBO loss.

HOW : Avg. 26 cases/month reported at Buyoff and 6-7 cases at LSP.

Noise while glass up/ down

- Noise observed while glass up/down

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will…

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

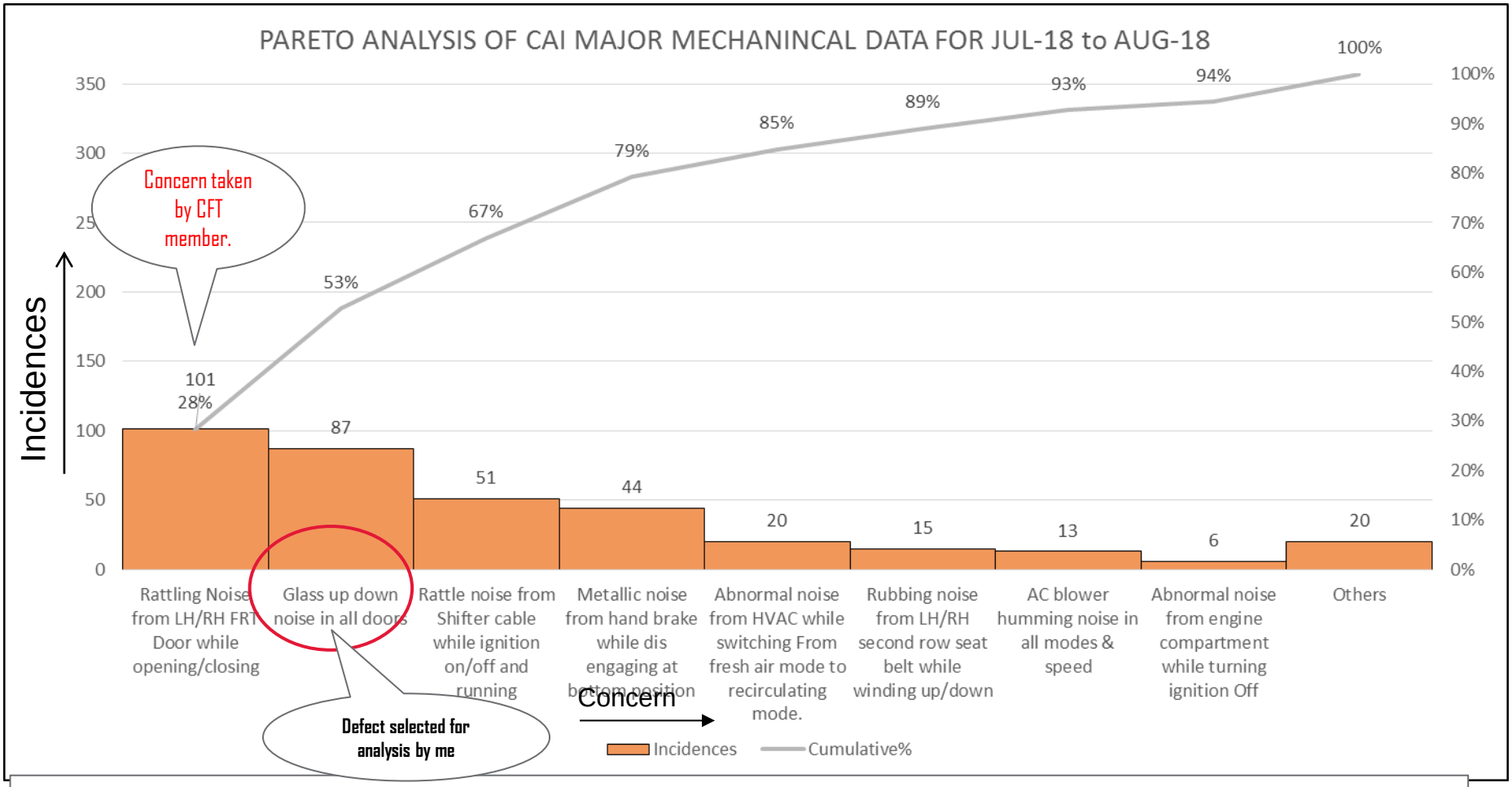
We will…

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for en[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 ye[illegible]
- Reduce carbon & water foo[illegible]nt by 25%

we
nd.

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Inference:

Glass up down noise in all doors it is 2nd top concern and repeated chronic concern, hence taken for resolution

2. Justify the choice

Severity & Consequences of Concern :-

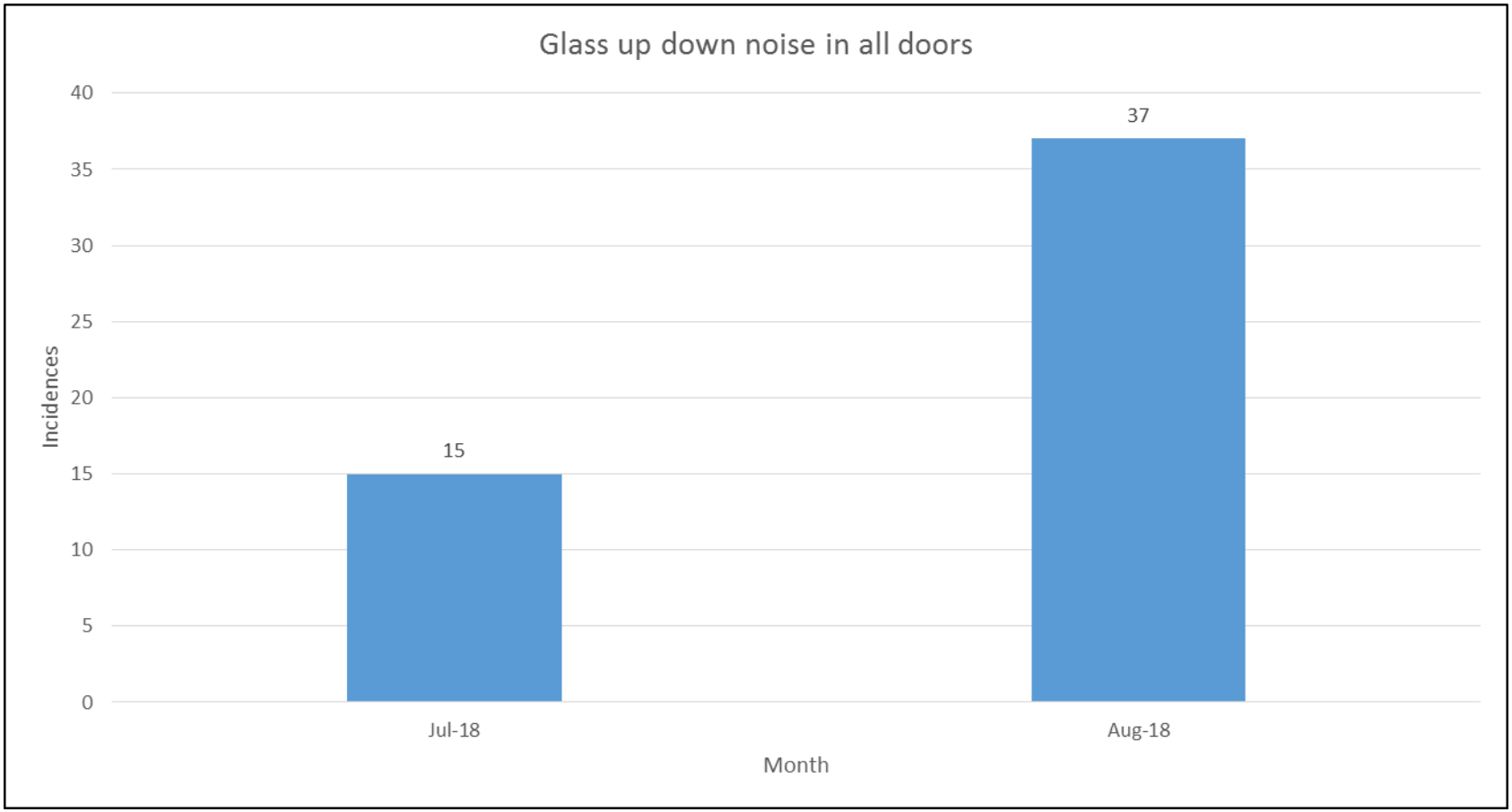
- **Avg. 26 Cases reported per month, for rework, 2 V1 concern reported in U321_LSP audit.**
- **Rework difficult**
- **Rework cost more.**
- **Rework time 7 min (approx.) for each vehicle by 1 operator.**
- **Rework cost is 11922 Rs/year**

Cost of Poor Quality - Rework

Rework Cost Estimation		
Description	UoM	
Rework Time	Min	7
Avg concerns/Day	Nos	8
Total rework time required per day	Min	56
Effective working time	Min	465
Operator required /day	Nos	0.12
Approx operator cost/day	Rs	330
Operator cost/concern	Rs	39.74
Avg no of working days	Days	25
Operator cost /month	Rs	993.55
Operator cost/Year	Rs	11922.58

2. Justify the choice

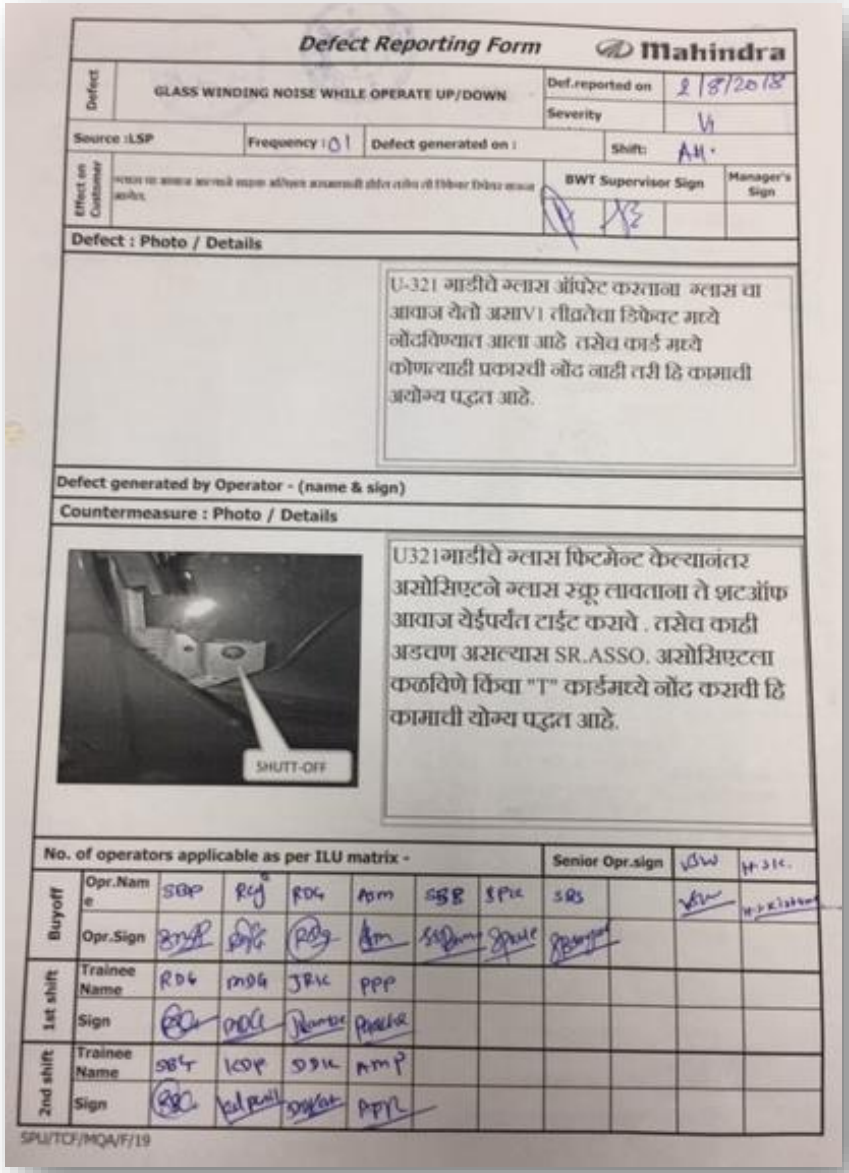
MONTHWISE DEFECT TREND :-



Inference : Average 26 No.s concerns reported per month from Jul -18 to Aug-18

2.b Immediate Containment Action

- Training for stage associates, explaining main step and key points.
- Emphasis on glass screw loose fitment as shown.
- 100% lot checked for supplier variability.
- DRF given to stage associates, explaining key points and focus on glass screw loose & glass up down noise.



Defect Reporting Form — Mahindra

Defect	GLASS WINDING NOISE WHILE OPERATE UP/DOWN	Def.reported on	2/8/2018
		Severity	4
Source : LSP	Frequency : 01	Defect generated on :	Shift: AM
Effect on Customer	[illegible]	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

U-321 गाडीचे ग्लास ऑपरेट करताना ग्लास चा आवाज येतो असा V1 तीव्रतेचा डिफेक्ट मध्ये नोंदविण्यात आला आहे तसेच कार्ड मध्ये कोणत्याही प्रकारची नोंद नाही तरी हि कामाची अयोग्य पद्धत आहे.

Defect generated by Operator - (name & sign)

Countermeasure : Photo / Details

U321गाडीचे ग्लास फिटमेन्ट केल्यानंतर असोसिएटने ग्लास स्क्रू लावताना ते शटऑफ आवाज येईपर्यंत टाईट करावे . तसेच काही अडचण असल्यास SR.ASSO. असोसिएटला कळविणे किंवा "T" कार्डमध्ये नोंद करावी हि कामाची योग्य पद्धत आहे.

No. of operators applicable as per ILU matrix - | Senior Opr.sign

Buyoff	Opr.Name	SOP	RG	RDG	AM	SSB	SPK	SRS			
	Opr.Sign										
1st shift	Trainee Name	RDG	MDG	JRK	PPP						
	Sign										
2nd shift	Trainee Name	SBG	KOP	SDK	AMP						
	Sign										

SPU/TOF/MQA/F/19

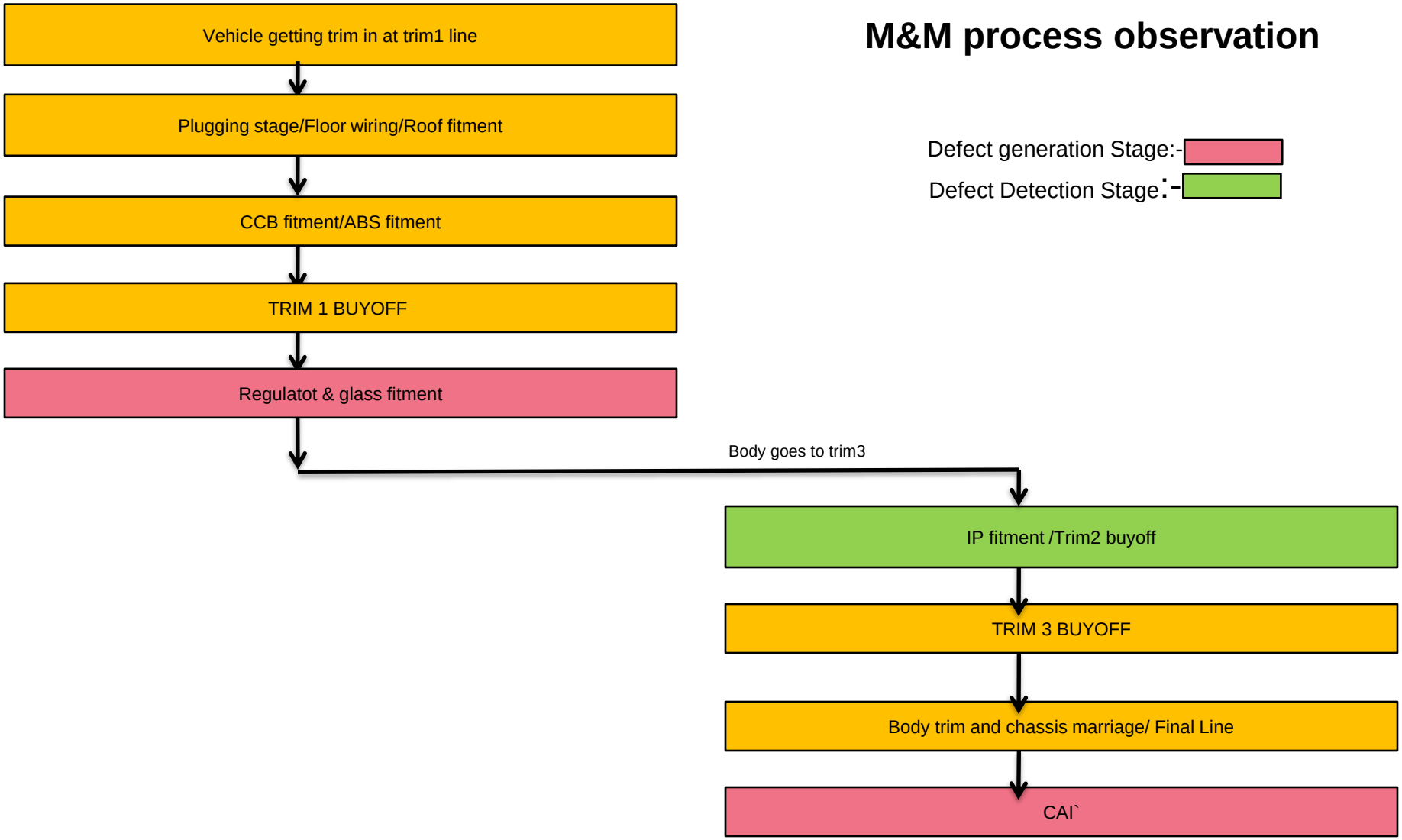
3. Understand the current situation

Flow Chart :- Observation of Process

M&M process observation

Defect generation Stage:-

Defect Detection Stage:-



Step 3. Understanding Current Situation

Observation of process

Door run channel fitment

Regulator fitment

Glass fitment

Triangular patch fitment

Glass up down movement checked at stage

Glass up down noise came from all doors.

CHECKING AT TRIM2 BUYOFF

Defect generation stage
Defect identification stage

Step 3. Understanding Current Situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area :STCF
Cell / BWT / Line:Trim 1/BWT3

Requested By- MQA | Date -17.09.18 | Source -Field | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement :Power Window Operating noise LH FR- Glass mounting screw loose

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- 18141 | Model :Marazzo

Date of Incidence/ Assy : 03/07/18 | Shift / Time : 1st | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
Yard check done

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☐ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - Dinesh Nagrale | BWT . : 03

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Operator Name : D D Khairnar

Man(Human Action)

Is operator trained ?	☑ YES ☐ NO
Operator level (I.L.U)	I
OR Dexterity level(1...4)	3
Is operator under training ?	☐ YES ☑ NO
Is operator temporary for Stage ?	☐ YES ☑ NO
Bad rework ?	☐ YES ☑ NO

Method(process)

Main steps are respected ?	☑ YES ☐ NO
Is there a related key point in SOP / SOS?	☑ YES ☐ NO
Are the reasons for key point known ?	☑ YES ☐ NO
Are operating reqmnts respected ?	☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ?	☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ?S.B.Pawar	☐ YES ☑ NO

Other

Is there a Control Plan ? (Record in Analysis table)	☐ YES ☐ NO
Is the packaging standard defined for the part ?	☐ YES ☐ NO
Is the packaging standard respected ?	☐ YES ☐ NO
Is there any alert mechanism for failure ?	☐ YES ☐ NO

Machine

Is the specified /Tool / Fixture facility used ?	☐ YES ☐ NO
Last calibration dates	
Facility /Tool / Fixture in good condition ?	☑ YES ☐ NO
Is it a substitute facility ?	☐ YES ☐ NO
Are machine parameters known ? (Record in Analysis table)	☐ YES ☑ NO
Parameters are respected ?	☐ YES ☐ NO

Product Failure

In only one shift ?	☑ YES ☐ NO
In only one line ?	☑ YES ☐ NO
The problem follows the vehicle ?	☑ YES ☐ NO
The problem follows the part ?	☐ YES ☑ NO
Is it a repeat problem ?	☐ YES ☑ NO
Is the suspected parameter measurable ? (Record in Analysis table)	☐ YES ☐ NO

Material/ Consumable

Is the specified product used ?	☐ YES ☐ NO
Did it evolve / Change ?	☐ YES ☐ NO
Is the storage SOP respected ?	☐ YES ☐ NO
Is the shelf life / expiry date respected ?	☐ YES ☐ NO
Is the part as per FIFO?	☐ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
After analysis found that glass screw loose.
On Dtd-3rd July 2018 in 1st shift Mr D D Khairnar (Black L)deployed for Fr door glass fitment.He has done the fitment as per SOS.We have some observation as below.
We have checked residual torque of the screw & it was found between 1.015 to 1.35 Nm.The residual torque was found less & we have analyzed the case.
☐ On line we are giving 1.9Nm torque which is common between Scorpio & U-321.
☐ The screw & clip are both common.
☐ Still the concern is there in U-321
The screw which we are using is M6 and the clearance in U-321 is 0.5mm where as in Scorpio it is 0.25mm.

Sno	Product / Process / Machine / Parameter	U321	Scorpio	Remark
1	Hole diameter	7.1 mm	6.5 mm	The hole diameter is more by 0.6mm.
2	Slot width	7.1 mm	6.5 mm	The slot width is more by 0.6mm
3	Slot length	10.8 mm	10.5 mm	The slot length is more by 0. 3 mm

Is 4M Analysis Part 2 required ? ☐ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☐ NO | **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action
1.Awareness given to stage associate through DRF.
2.Torque set on higher side tolerance i.e.2.1NM

Resp	Date	Status
Dinesh Mangesh M	17.09.18	Close

2. Permanent Action -
L1 required.Further action required from design dept

Resp	Date	Status
Mangesh M		

Analysed by Name/s - Dinesh Nagrale
Date - 17.09.18

Choice : ☐ WSS ☐ QC Story

L0 ANALYSIS FOR POOR FIT

Inference from L0
Problem follows part & vehicle
Problem is a repeat problem
L1 analysis required

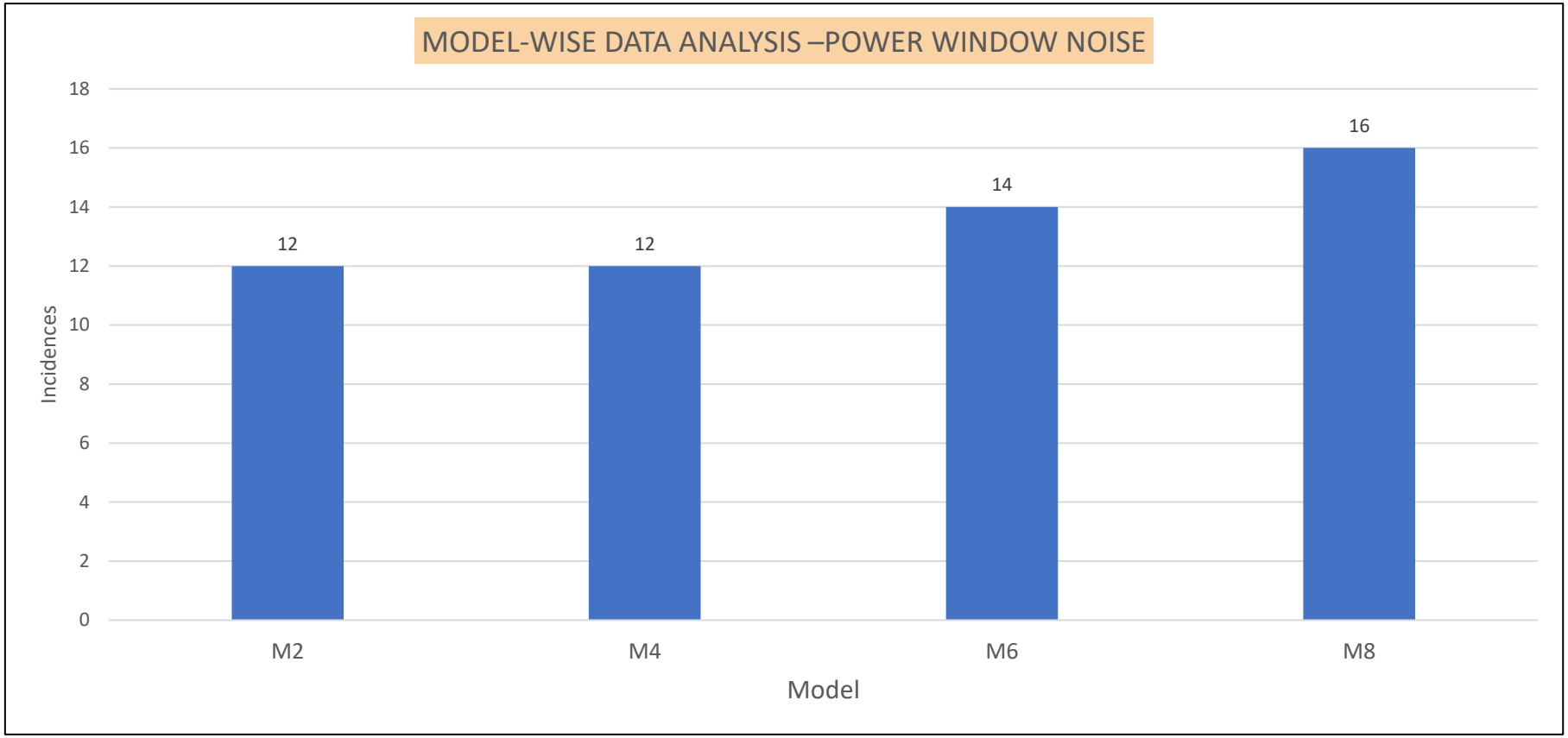
4M analysis Good and Not so good observations

MAN	Operator dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part and vehicle
Material	Handling and shelf life of part adhered

NO QUICK WIN

Step 3. Understanding Current Situation

MONITORING OF CONCERN-POWER WINDOW NOISE :



INFERENCE:-

Defect observed in all variant of Marazzo model. Defect is not model specific.

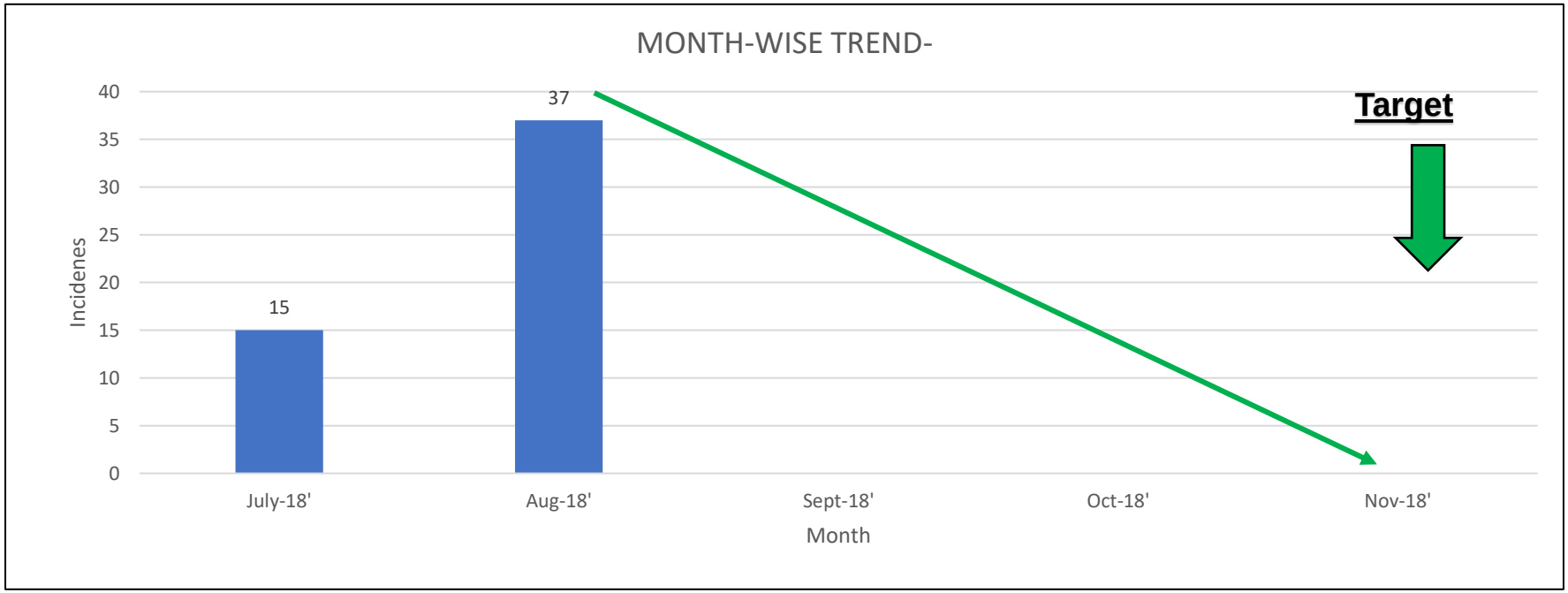
Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed in Marazzo all Model.
- Issue reported in both the shifts
- Repair time is high, around 7 minutes per vehicle
- Level 3 operators are deployed for regulator & glass fitment.
- Operators are aware of key point & check point at respective stages.
- Power Window Operating noise for all doors, Glass mounting screw loose in Marazzo all vehicles.

4. Select targets

Present Level:	**Avg. 26 Incidences are being reported at Trim 3 buy off.**
Target Level:	**Eliminate concern Glass up down noise for all doors in Marazzo of Nov -18.**



Step 5. Analysis

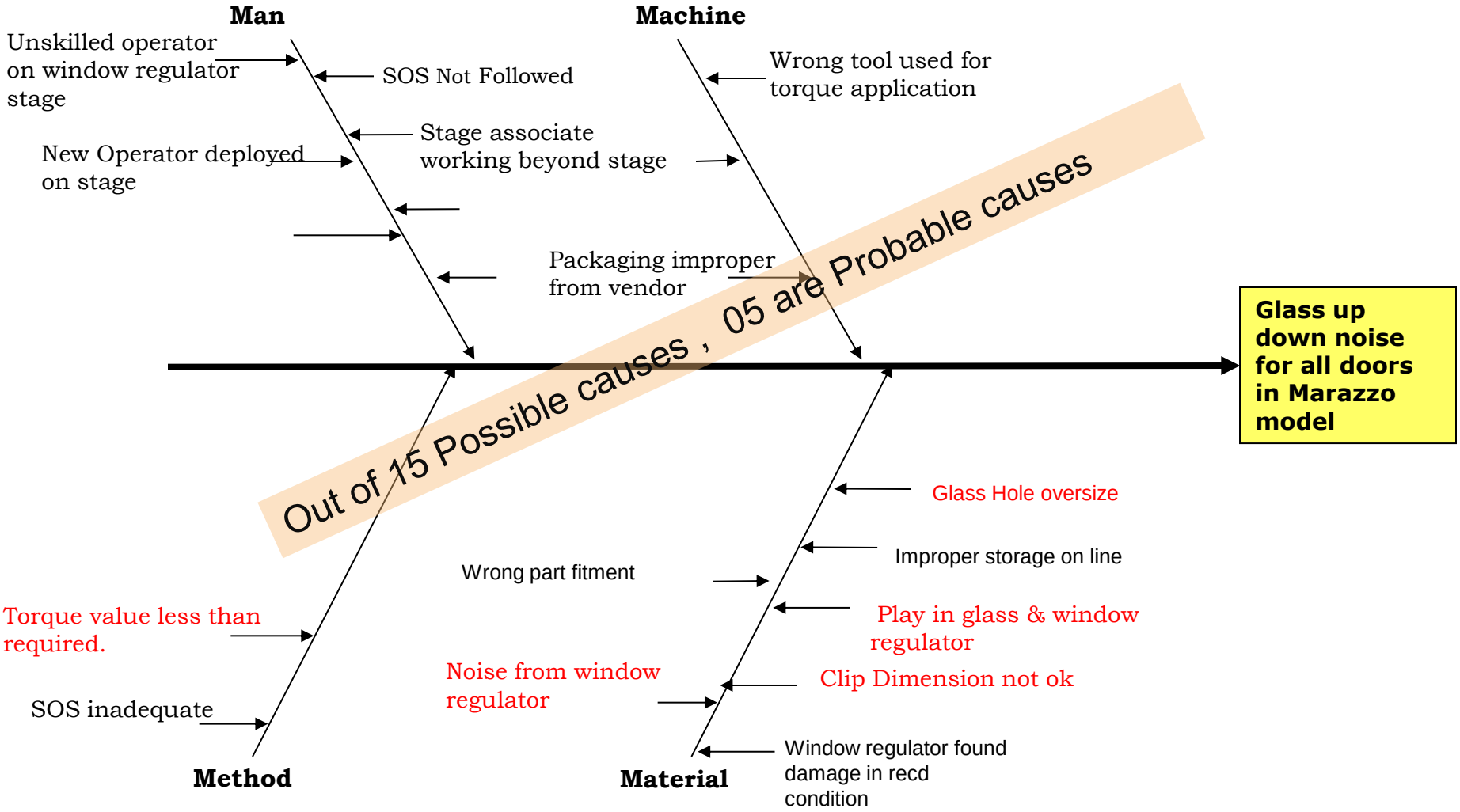
List of Possible causes identified in Brain storming session

Brainstorming Done along with CFT & we have found out 20 Possible Causes for this defect.

Sr. no	**Possible Causes**
1	New operator deployed at stage.
2	SOS Not Followed.
3	Unskilled operator for window reglator fitment
4	SOS Inadequete
5	Window regulator damage in recived condition
6	Wrong tool used for torque application
7	Torque value less than rerquired.
8	Play in glass & window regulator
9	Clip Dimention not ok
10	Glass Hole oversize
11	Noise from window regulator
12	Wrong part fitment
13	Stage associate working beyong stage
14	Improper storage on line
15	Packaging impropoer from vendor

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes :

Sr No	Possible causes	Prasad	Satish	Amar	Mahesh	Ganesh	Mangesh	Total
1	New operator deployed at stage	1	1	3	3	3	3	14
2	SOS Not Followed	1	1	1	1	1	1	6
3	Unskilled operator for window reglator fitment	3	3	3	3	1	9	22
4	SOS Inadequete	1	3	1	1	1	3	10
5	Window regulator damage in recived condition	1	1	1	1	1	1	6
6	Wrong tool used for torque application	1	1	3	3	1	3	12
7	Torque value less than rerquired.	3	3	9	9	9	3	36
8	Play in glass & window regulator	9	9	9	9	3	9	48
9	Clip Dimention not ok	9	9	9	9	3	3	42
10	Glass Hole oversize	9	9	9	9	3	9	48
11	Noise from window regulator	9	9	9	3	9	3	42
12	Wrong part fitment	0	0	0	0	1	0	1
13	Improper storage on line	1	1	0	0	0	3	5
14	Packaging impropoer from vendor	3	1	3	1	0	3	11
15	Stage associate working beyong stage	3	1	3	3	3	3	16

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 35.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering.

Sr No	Possible causes	Reasons
1	New operator deployed at stage	All trained operator deployed on stage
2	SOS Not Followed	SOS followed
3	Unskilled operator for window reglator fitment	All trained operator deployed on stage
4	SOS Inadequete	Check & found OK
5	Window regulator damage in recived condition	Check & found OK
6	Wrong tool used for torque application	Visually check confirmed Ok
12	Wrong part fitment	Correct oart used.
13	Improper storage on line	Checked & found Ok
14	Packaging impropoer from vendor	Storage condition is audited & found ok
15	Stage associate working beyong stage	Stage condition audited & found Ok

5.1 Identify Probable cause

Sr No	Possible causes
7	Torque value less than rerquired.
8	Play in glass & window regulator
9	Clip Dimention not ok
10	Glass Hole oversize
11	Noise from window regulator

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

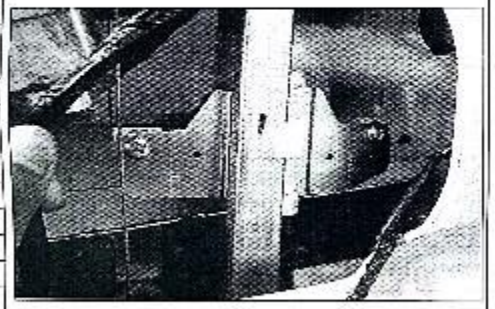
No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Torque value less than required.	Residual torque value checked & torque value verification in torque chart	Total=25veh checked for residual torque, torque value found within spec of 1.9±0.1Nm. Torque chart verified & torque spec found OK. Still noise observed in 07vehicles.	**Hypothesis valid**

CHECKSHEET

Desc : Glass up down noise from power window regulator — Date : 10.07.2018

Model : W115

Sr NO	CHECKPOINTS	WHAT TO CHECK	HOW TO CHECK	Observation \Result
1	Residual Torque	Taoruc Spec 1.9±0.1Nm	Torque checking by ACTA	1·91
2				1·90
3				1·91
4				1·89
5				1·88
6				1·91
7				1·90
8				1·89
9				1·91
10				1·88
11				1·90
12				1·91
13				1·89
14				1·90
15				1·89
16				1·91
17				1·90
18				1·91
19				1·89
20				1·88



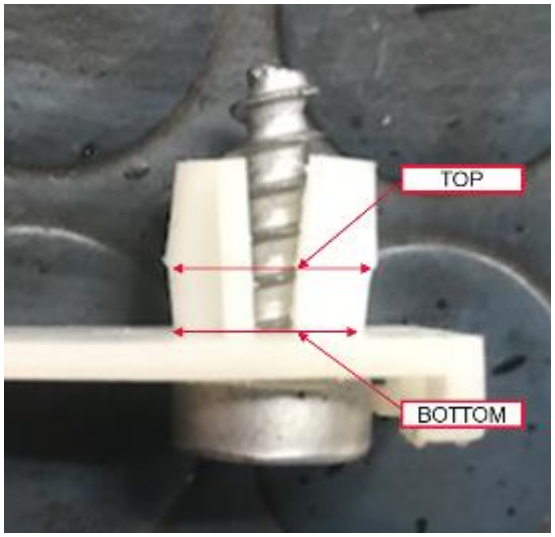
Torque value found within spec , still noise observed during glass up down. Also play observed in glass & regulator mtg.

Inference:- Detail joint analysis required with Torque study

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Play in glass & window regulator	Visual inspection & dimensional analysis	Total 10vehicles checked play found in all vehicle. From dimensional analysis its observed that interference between glass hole & mtg clip is 0.4mm only.	**Hypothesis valid**



The clip got flared & it was observed 11.8 mm at bottom, where as glass dia was 11.39 mm. **The interference at this point is 0.4mm which is very less**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

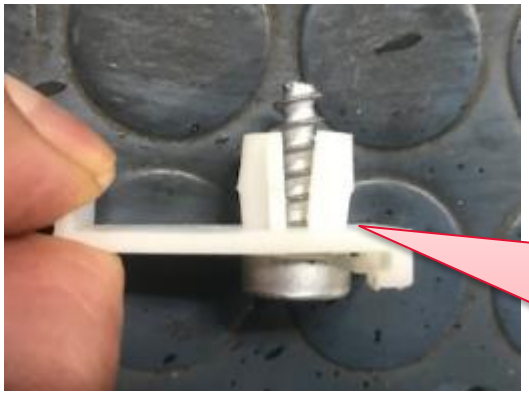
Comparison Study between Scorpio & U321:-

- We have compared the parameters of both Scorpio & U321
- Except glass hole dia & Screw pitch all other parameters are same.

Sr No	Characteristic	Specification Scorpio	Specification U321
1	Glass Hole Diameter	11.5 +/- 0.5mm	11.0 +/- 0.5mm
2	Mounting clip step	11.0+/- 0.2 mm	11.0+/- 0.2 mm
3	Screw major dia	6.0 to 6.24 mm	6.0 to 6.24 mm
4	Screw Pitch	1.8	2.69
5	Screw Length	18.3	18.3

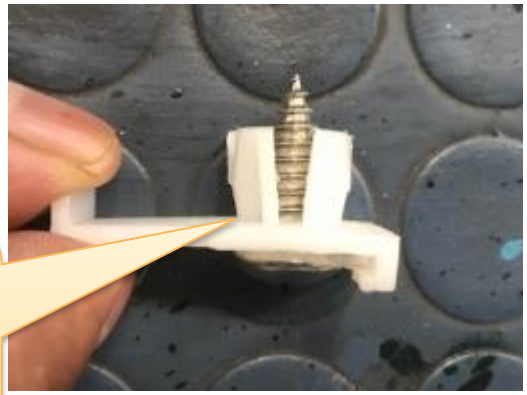
From Above data we have decided to study the inference of clip & hole after the screw tightening

- When the screw is inserted the clip gets flared and the glass hole mounting clearance gets eliminated.
- After inserting the screw the flaring dimensions were found different for both screws as shown below



The clip got flared & it was observed 11.8 mm Where as glass dia was 11.39 mm. **The interference at this point is 0.4mm**

The clip got flared & it was observed 12.96 mm Where as glass dia was 11.9 mm. **The interference at this point is 1.06 mm**



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Clip Dimension not ok	Visual check	Total 05 clips checked all dimensions found within spec.	**Hypothesis Invalid**

MAHINDRA & MAHINDRA LTD,
AUTOMOTIVE SECTOR,
NASHIK - 422 007.

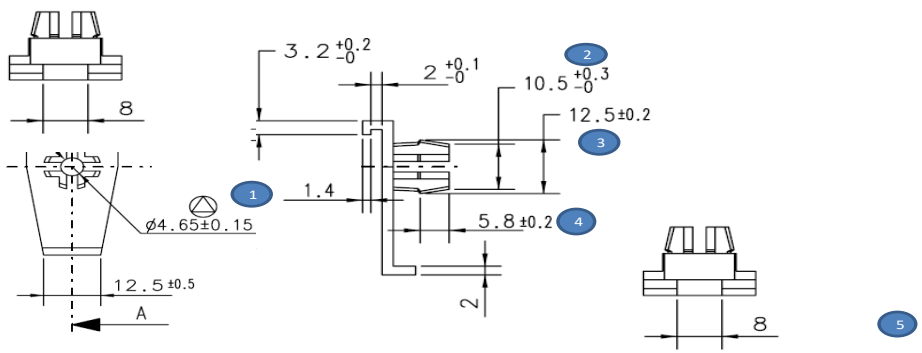
DEPT: MQA - PLANT QUALITY

Date : 25.08.2018
Page 1 of 1

INSPECTION REPORT

Part No : 0111CG0080N
Vendor Code :
Drg.CH. Let :

Desc : Clip
Name : Trim India Pvt ltd.
Ref.No :



Sr. No.	Product Charactaristics	Spec	UOM	Inspection Method	Actual Observations					Remark
1	Hole ID	4.65±0.15	mm	vernier	4.68	4.68	4.66	4.68	4.66	Ok
2	Dimension	10.5+0.3	mm	vernier	10.62	10.6	10.62	10.6	10.62	Ok
3	Dimension	12.5±0.2	mm	Height Gauge	12.52	12.54	12.52	12.54	12.52	Ok
4	Dimension	5.8±0.2	mm	vernier	5.9	5.9	5.9	5.92	5.9	Ok
5	Dimension	8	mm	Vernier	7.92	7.94	7.92	7.94	7.92	Ok

Prepared By : Satish Ckecked by : Prasad

Part found within specification as per layout report.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

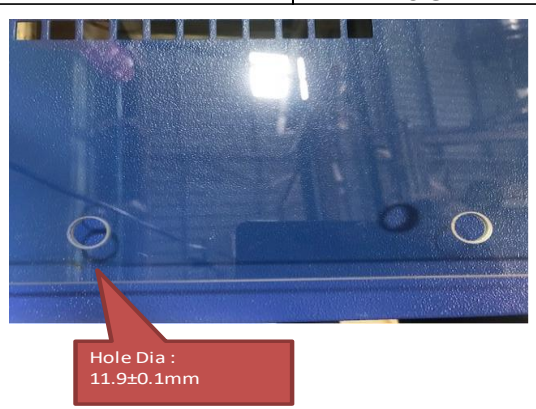
No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Glass Hole oversize	Visual check	Total 20 glass checked, hole dimension found within spec of 11±0.5mm.	**Hypothesis Invalid**

CHECKSHEET

Desc : Front Door glass — Date : 10.01.2016

Model : Marazzo

Sr NO	CHECKPOINTS	WHAT TO CHECK	HOW TO CHECK	Observation \Result
1	Glass hole dimention	Hole Dia : 11±0.5mm	Vernier Caliper	11.36
2				11.38
3				11.38
4				11.35
5				11.38
6				11.35
7				11.38
8				11.38
9				11.38
10				11.38
11				11.38
12				11.38
13				11.38
14				11.38
15				11.38
16				11.35
17				11.38
18				11.38
19				11.36
20				11.38



Part found within specification as per linspection report.

TEST OF HYPOTHESIS

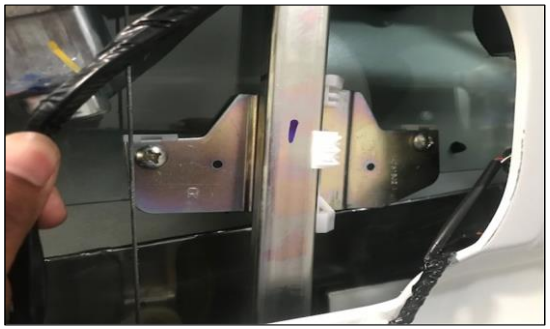
5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Noise from window regulator	Listen carefully while glass up down.	Total 20 vehicles checked, Noise not observed from window regulator.	**Hypothesis Invalid**

CHECKSHEET

Desc : Glass up down noise from power window regulator — Date : 10.07.2018

Model : MARAZZO

Sr NO	CHECKPOINTS	WHAT TO CHECK	HOW TO CHECK	Observation \Result
1	Window up down noise from regulator	Noise	Listen carefully while glass up down	No
2				No
3				No
4				No
5				No
6				No
7				No
8				No
9				No
10				No
11				No
12				No
13				No
14				No
15				No
16				No
17				No
18				No
19				No
20				No

No noise observed from window regulator in 20 checked vehicles hence hypothesis is invalid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Torque value less than required.	Torque inspection with ACTA & physical checking	**Hypothesis valid**
2	Play in glass & window regulator	Visual inspection & dimensional analysis	**Hypothesis valid**

2 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION / COUNTER MEASURE
1	Torque value less than required	Torque applied as per given specification	Torque value as per torque chart i,.e design.			**Torque value to be change in torque chart.**
2	Play in glass & window regulator	Clip not getting expand fully after specified torque application.	Screw not suitable for clip expansion in given hole size.	**Glass clip mtg screw pitch dimension not suitable for clip expansion.**	**Screw Pitch dimension incorrect in drawing**	**Drawing to revised & screw to be change as per required pitch value.**

6. Implement the corrective measures

Legends

↑ High ↓ Low

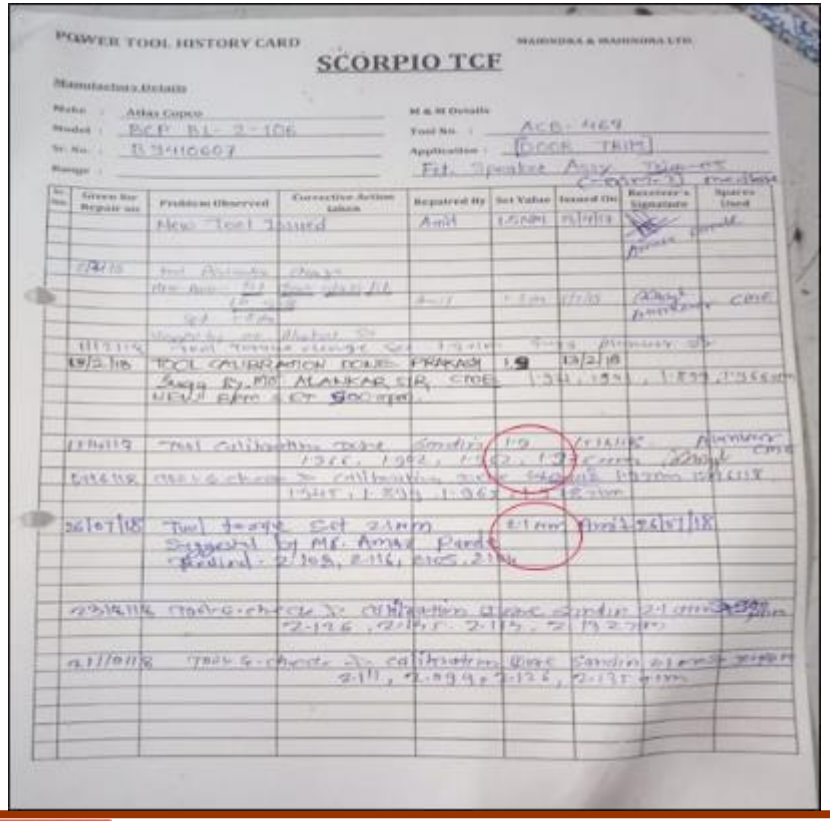
Possible Remedies Identified

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Torque value as per torque chart i,.e design.	**Torque value to be change in torque chart.**	Low ↓	Low ↓	Yes	Low ↓	**Accepted**
2	**Screw Pitch dimension in correct in drawing**	**Drawing to revised & screw to be change as per required pitch value.**	High ↑	Low ↓	Yes	Low ↓	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
I	Torque value as per torque chart i,.e design.	**New Torque value changed in torque chart.** **New torque value-2.1 Nm**	26.07.18	Satish Pagar	Closed

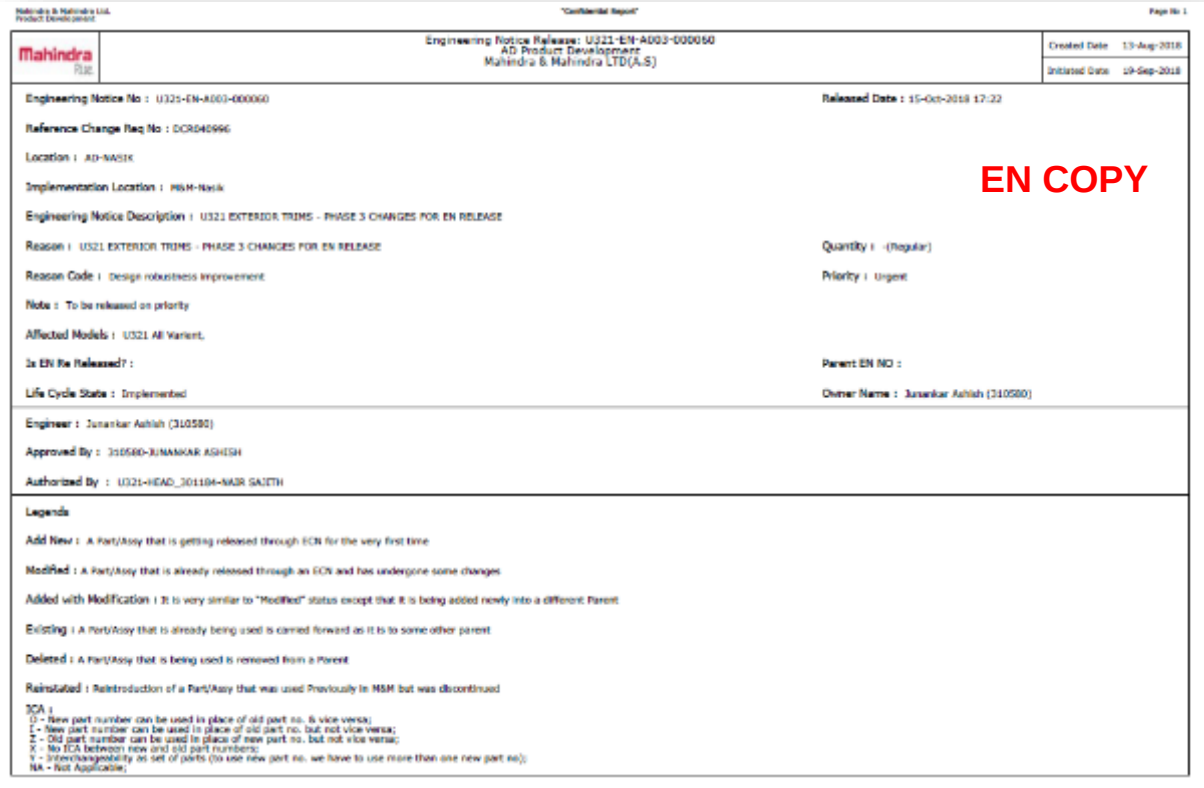


New torque value-2.1 Nm

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	Screw Pitch dimension not correct in drawing	Drawing revised & screw change as per new required pitch value.	09.08.2018	Rohan Kamat	Closed



Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra

Engineering Notice Release: U321-EN-A003-000060
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 13-Aug-2018
Initiated Date 19-Sep-2018

Engineering Notice No : U321-EN-A003-000060

Released Date : 15-Oct-2018 17:22

Reference Change Req No : DCR040996

Location : AD-NASIK

Implementation Location : M&M-Nasik

EN COPY

Engineering Notice Description : U321 EXTERIOR TRIMS - PHASE 3 CHANGES FOR EN RELEASE

Reason : U321 EXTERIOR TRIMS - PHASE 3 CHANGES FOR EN RELEASE

Quantity : -(Regular)

Reason Code : Design robustness improvement

Priority : Urgent

Note : To be released on priority

Affected Models : U321 All Varient.

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Implemented

Owner Name : Junankar Ashish (310580)

Engineer : Junankar Ashish (310580)

Approved By : 310580-JUNANKAR ASHISH

Authorized By : U321-HEAD_301104-NAIR SAJITH

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	Screw Pitch dimension not correct in drawing	Drawing to revised & screw to be change as per required pitch value.	09.08.2018	Rohan Kamat	Closed

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 3

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000060
AD Product Development
Mahindra & Mahindra LTD(A.S)

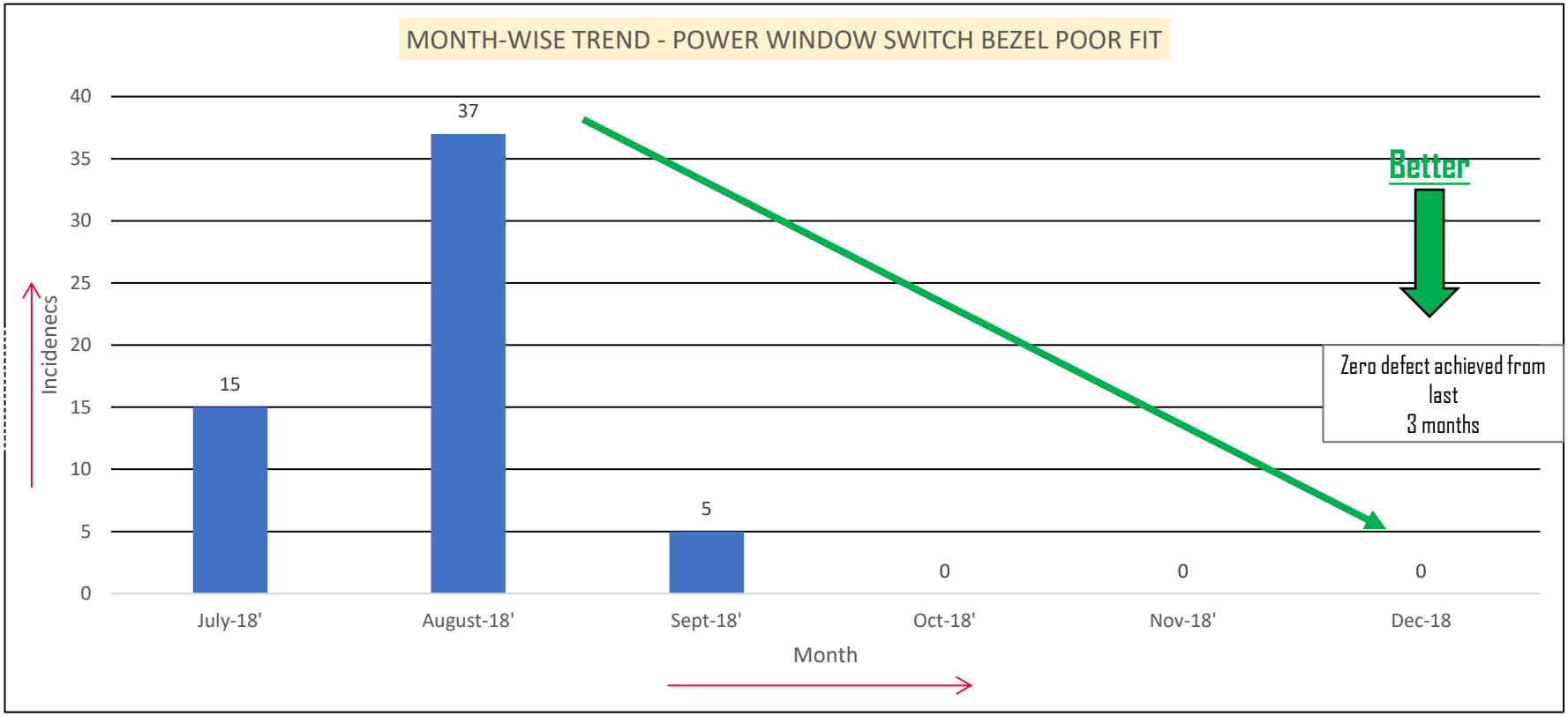
Created Date 13-Aug-2018
Initiated Date 19-Sep-2018

Part Number	Rev	Nomenclature	Status	ICA	Qty	Replaced/ Deleted Part	Parent Part	Variant Condition	Responsibility	Remarks	Discontinued
0119AU300450N	C/035	BUMPER ASSY COMP FRT HI PLUS	Modified	NA	1		0108EU300780N		RUPESH_A (23062683)		No
0108FU300160N	B/008	INSTALLATION ASSY - APPLIQUE ASSY D PILLAR LH	Modified	NA	1		0108EU300780N		RUPESH_A (23062683)		No
0119AU301050N	C/009	FASCIA ASSY FRONT UPPER BASE MID	Modified	NA	1		0119AU300800N		RUPESH_A (23062683)		No
0119AU300560N	C/031	FRONT BUMPER MAIN SUBASSY	Modified	NA	1		0119AU300450N		RUPESH_A (23062683)		No
2302AU300921N	B/002	U CLIP FRONT BMPR SIDE FLANGE	Modified	NA	2		0119AU301040N		RUPESH_A (23062683)		No
0108FU300040N	B/022	D PILLAR TRIM LH	Modified	NA	1		0108FU300080N		RUPESH_A (23062683)		No
SF0202114	A/001	SCREW HEX FL M6X1X18X8.8	Deleted	NA	4		0119BU300830N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	4	SF0202114	0119BU300830N		RUPESH_A (23062683)		No
0108EU300240N	C/017	APPLIQUE ASSY RR DR LH	Modified	NA	1		0108EU300390N		RUPESH_A (23062683)		No
0108EU300210N	B/035	ROCKER MOULDING SINGLE PNL RH	Modified	NA	1		0108EU300220N		RUPESH_A (23062683)		No
0102DU300120N	B/035	SPLASH SHIELD FRONT RH FRT	Modified	NA	1		0102DU300100N		RUPESH_A (23062683)		No
0119BU300860N	C/011	REAR BUMPER SUBASSY BASE MID	Modified	NA	1		0119BU300450N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	4	SF0202133	0119BU300450N		RUPESH_A (23062683)		No
0108EU300780N	C/020	TOP LEVEL BODY EXTERIOR ASSY	Modified	NA	1		0000AU30030LA		RUPESH_A (23062683)		No
0108EU300180N	B/036	ROCKER MOULDING SINGLE PNL LH	Modified	NA	1		0108EU300190N		RUPESH_A (23062683)		No
SF0202018	A/001	SCREW HEX FL M6X1X20X8.8XGR	Deleted	NA	1		0108FU300170N		RUPESH_A (23062683)		No
0108FU300090N	B/014	D PILLAR ASSY TRIM RH	Modified	NA	1		0108FU300170N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	1	SF0202018	0108FU300170N		RUPESH_A (23062683)		No

ADD DELETE ENTRY IN EN COPY FOR NEW SCREW

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – Concern not reported from after implementing the PCA actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	EN Change/drawing updation	09.08.18	Rohan Kamat	CLOSED
2	BOM Updation	10.08.18	Vikrant Morade	CLOSED
4	Tool history card updation	15.09.18	Satish Pagar	CLOSED
5	PFMEA updation (Rev No –02)	20.08.18	Amar Pande	CLOSED
6	Process audit check sheet updation (Rev No -6)	10.09.18	Satish Pagar	CLOSED
7	SOS Updation (Rev No - 02)	25.09.18	Prasad Deore	CLOSED
8	On job Training given to stage associates for power window switch bezel fitment.	25.08.18	Prasad Deore	Closed

Check & confirm the results & effects (Cont……)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of concern glass up down noise.
2) FRC/FBO improved by 0.1%.
3) Rework cost save upto11922 Rs / Year

In tangible Benefits

1) Improved Customer Satisfaction.
2) Operator fatigue reduced , moral improved
3) Vehicle aesthetic improved.

From: DHANOKAR RAJESH
Sent: Friday, December 01, 2017 11:58 AM
To: DEORE PRASAD-NASHIK
Subject: RE: Rework cost Estimation

Rework Cost Estimation

Description	UoM	
Rework Time	Min	7
Avg concerns/Day	Nos	8
Total rework time required per day	Min	56
Effective working time	Min	465
Operator required /day	Nos	0.12
Approx operator cost/day	Rs	330
Operator cost/concern	Rs	39.74
Avg no of working days	Days	25
Operator cost /month	Rs	993.55
Operator cost/Year	Rs	11922.58

Cost Approval mail from Industrial engineering

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S N	Document	Target date	Responsibility	Status
1	EN Change/drawing updation	25.09.18	Rohan Kamat	Closed
2	BOM Updation	15.09.18	Vikrant Morade	Closed
3	Tool history card updation	15.09.18	Satish Pagar	Closed
4	PFMEA updation (Rev No –02)	20.08.18	Amar Pande	Closed
5	SOS Updation (Rev No - 02)	25.08.18	Prasad Deore	CLOSED
6	QA matrix updated	06.08.2018	Prasad Deore	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	One point lesson	25.08.18	Prasad Deore	Closed

8.3 Additional Audit & check :

SN	Document	Target date	Responsibility	Status
1.	Process Audit check sheet	10.09.18	Satish Pagar	CLOSED

8. Standardization

8.1 Updation of documents : -

S N	Document	Target date	Responsibility	Status
1	EN Change/drawing updation	25.09.18	Rohan Kamat	Closed

Engineering Change Note :

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

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Mahindra Rise

Engineering Notice Release: U321-EN-A003-000060
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 13-Aug-2018
Initiated Date 19-Sep-2018

Engineering Notice No : U321-EN-A003-000060

Released Date : 15-Oct-2018 17:22

Reference Change Req No : DCR040996

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : U321 EXTERIOR TRIMS - PHASE 3 CHANGES FOR EN RELEASE

Reason : U321 EXTERIOR TRIMS - PHASE 3 CHANGES FOR EN RELEASE

Quantity : -(Regular)

Reason Code : Design robustness improvement

Priority : Urgent

Note : To be released on priority

Affected Models : U321 All Variant,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Implemented

Owner Name : Junankar Ashish (310580)

Engineer : Junankar Ashish (310580)

Approved By : 310580-JUNANKAR ASHISH

Authorized By : U321-HEAD_301184-NAIR SAJITH

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

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Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

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I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

8. Standardization

8.1 Updation of documents : -

S N	Document	Target date	Responsibility	Status
2	BOM Updation	15.09.18	Vikrant Morade	Closed

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

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Mahindra Rise

Engineering Notice Release: U321-EN-A003-000060
AD Product Development
Mahindra & Mahindra LTD(A.S)

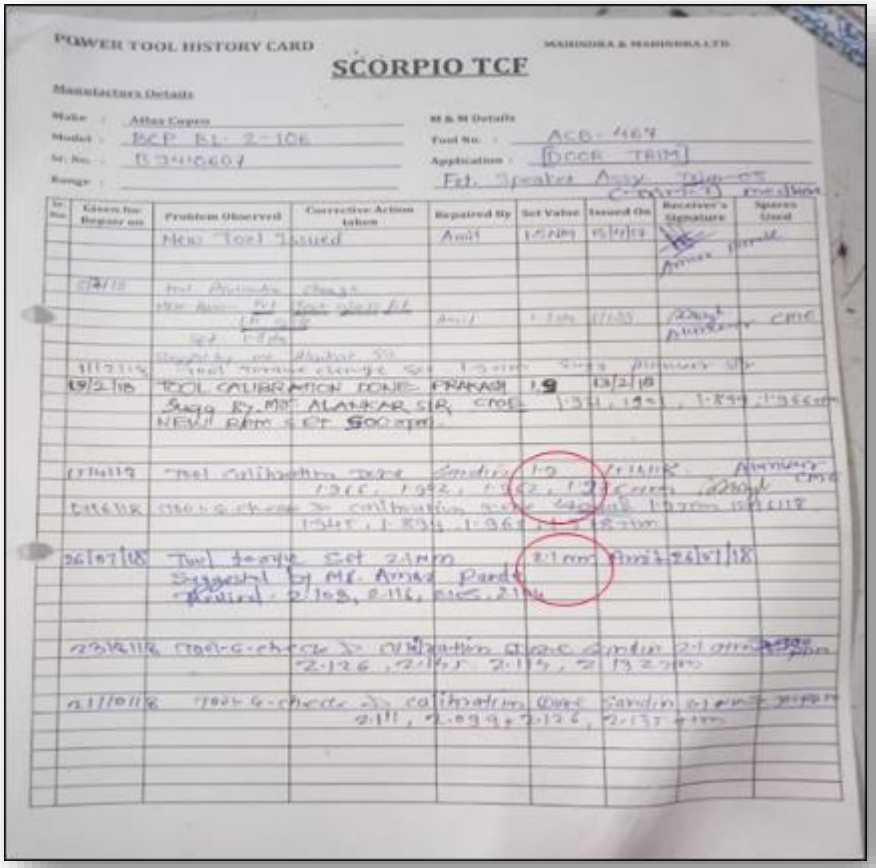
Created Date 13-Aug-2018
Initiated Date 19-Sep-2018

Part Number	Rev	Nomenclature	Status	ICA	Qty	Replaced/ Deleted Part	Parent Part	Variant Condition	Responsibility	Remarks	Discontinued
0119AU300450N	C/035	BUMPER ASSY COMP FRT HI PLUS	Modified	NA	1		0108EU300780N		RUPESH_A (23062683)		No
0108FU300160N	B/008	INSTALLATION ASSY - APPLIQUE ASSY D PILLAR LH	Modified	NA	1		0108EU300780N		RUPESH_A (23062683)		No
0119AU301050N	C/009	FASCIA ASSY FRONT UPPER BASE MID	Modified	NA	1		0119AU300800N		RUPESH_A (23062683)		No
0119AU300560N	C/031	FRONT BUMPER MAIN SUBASSY	Modified	NA	1		0119AU300450N		RUPESH_A (23062683)		No
2302AU300921N	B/002	U CLIP FRONT BMPR SIDE FLANGE	Modified	NA	2		0119AU301040N		RUPESH_A (23062683)		No
0108FU300040N	B/022	D PILLAR TRIM LH	Modified	NA	1		0108FU300080N		RUPESH_A (23062683)		No
SF0202114	A/001	SCREW HEX FL M6X1X18X8.8	Deleted	NA	4		0119BU300830N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	4	SF0202114	0119BU300830N		RUPESH_A (23062683)		No
0108EU300240N	C/017	APPLIQUE ASSY RR DR LH	Modified	NA	1		0108EU300390N		RUPESH_A (23062683)		No
0108EU300210N	B/035	ROCKER MOULDING SINGLE PNL RH	Modified	NA	1		0108EU300220N		RUPESH_A (23062683)		No
0102DU300120N	B/035	SPLASH SHIELD FRONT RH FRT	Modified	NA	1		0102DU300100N		RUPESH_A (23062683)		No
0119BU300860N	C/011	REAR BUMPER SUBASSY BASE MID	Modified	NA	1		0119BU300450N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	4	SF0202133	0119BU300450N		RUPESH_A (23062683)		No
0108EU300780N	C/020	TOP LEVEL BODY EXTERIOR ASSY	Modified	NA	1		0000AU30030LA		RUPESH_A (23062683)		No
0108EU300180N	B/036	ROCKER MOULDING SINGLE PNL LH	Modified	NA	1		0108EU300190N		RUPESH_A (23062683)		No
SF0202018	A/001	SCREW HEX FL M6X1X20X8.8XGR	Deleted	NA	1		0108FU300170N		RUPESH_A (23062683)		No
0108FU300090N	B/014	D PILLAR ASSY TRIM RH	Modified	NA	1		0108FU300170N		RUPESH_A (23062683)		No
SF0202220	A/001	SCREW HEX FL SL M6X1X20X9.8	Add Existing	I	1	SF0202018	0108FU300170N		RUPESH_A (23062683)		No

8. Standardization

8.1 Updation of documents : -

S N	Document	Target date	Responsibility	Status
3	Tool history card updation	15.09.18	Satish Pagar	Closed



8. Standardization

8.1 Updation of documents : -

S N	Document	Target date	Responsibility	Status
4	PFMEA updation (Rev No –02)	20.08.18	Amar Pande	Closed

Part : A800 | Model :Marazzo
Part Nom.: | Part Name/ Process: FRONT DOOR GLASS RH
Latest Change Letter: | Opts No: 111039
Pdt Shop :Scorpio, Marazzo TCF | Opts Despts: FRONT DOOR GLASS RH FITMENT
Modular Cell :TRIM LINE | MC No.:

Mahindra & Mahindra Ltd.

Process Potential Failure Mode & Effect Analysis.

Reviewed by :Amar Pande | Rev. No : 4
Approved by : Nandar John | Rev. Date : 09.09.18
Doc No: MQS/PF/MEJW2 MHO/03 PFCF/01111039
PMEA Date (Orig): 08.01.2014

Cross Functional Team: Vikrant Morade; Yogesh Dhawle; Vikrant Ghule; Sumit Sisodia

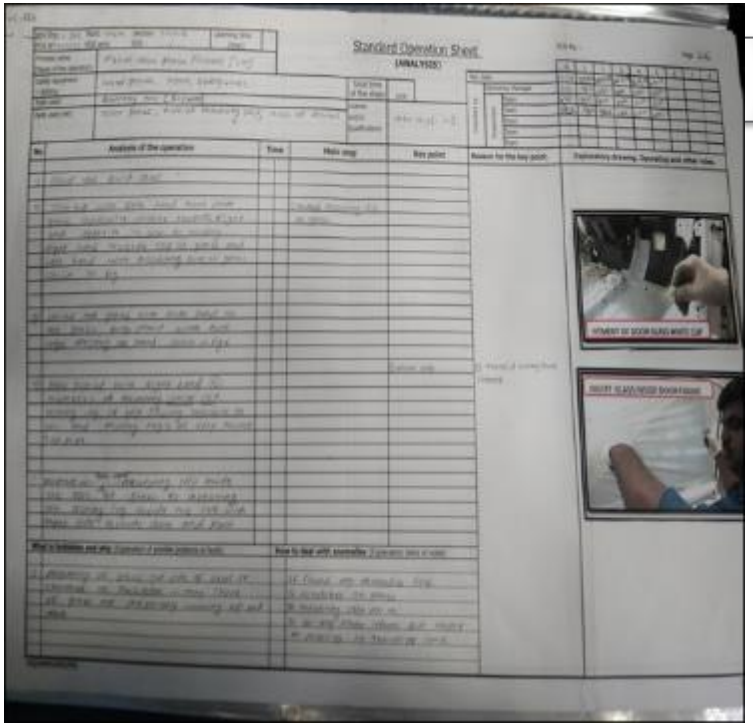
Process Step Function	Requirement	Potential Failure Mode	Potential Effects of Failures	S e v	C l a s s	Potential Causes/Mechanisms of Failures	Current Process Controls Prevention	O c c u	Current Process Controls Detection	D e t	R P N	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	S e v	O c c	D e t	R P N
SELECT SCREWS & POSITION ON GLASS MOUNTING CUPS	SELECTION OF SCREW AS PER MODEL	WRONG PART SELECTION	End customer:- customer dissatisfaction(Rattle noise) Internal customer:- Glass operating slow. Rework on next stage	4		OPERATOR MISTAKE	OPERATOR TRAINING	3	SELF CHECK			Screw to be common with scorpio	Dinesh N Sept-18	Action is closed				
SELECT TOOL & TIGHTEN TO SPECIFIED TORQUE	PROPER TIGHTENING OF THE JOINT	TORQUE NOT ACHIEVED-UNDER TORQUE (Field concern Sept-18)	End customer:- customer dissatisfaction(Rattle noise) Internal customer:- Glass operating slow. Rework on next stage	8		CALIBRATION MAY NOT HAVE DONE FOR THE REGULAR INTERVAL	(1)TORQUE CALIBRATION FREQUENCY TO BE DECIDE (2)HISTORY CARD OF THE TOOL TO BE MAINTAINED	2	AS PER AMC TOOL WILL BE CALIBRATED IN THE REGULAR INTERVALS									
						CALIBRATION SET VALUE IS WRONG	IF THE IDENTIFICATION LABEL IS MISSING NUTRUNNER SHOULD NOT BE USED	3	CALIBRATION RECORD			Tightening torque is to be increased to 2.1Nm from 1.9Nm	Dinesh N Sept-18	Action is closed				
						WRONG TOOL USED	DEDICATED TOOLS FOR SPECIFIED TORQUE AND LABELS ON THE TOOL	2	LABELS ON THE TOOL									
		TORQUE NOT ACHIEVED - OVER TORQUE	End customer:- customer dissatisfaction(thread damaged/slip, rattle noise) Internal customer:- glass operating slow. Rework on next stage	8		CLIBRATION SET VALUE IS WRONG	IF THE IDENTIFICATION LABEL IS MISSING NUTRUNNER SHOULD NOT BE USED	3	CALIBRATION RECORD									
						WRONG TOOL USED	DEDICATED TOOLS FOR SPECIFIED TORQUE AND LABELS ON THE TOOL	2	LABELS ON THE TOOL									

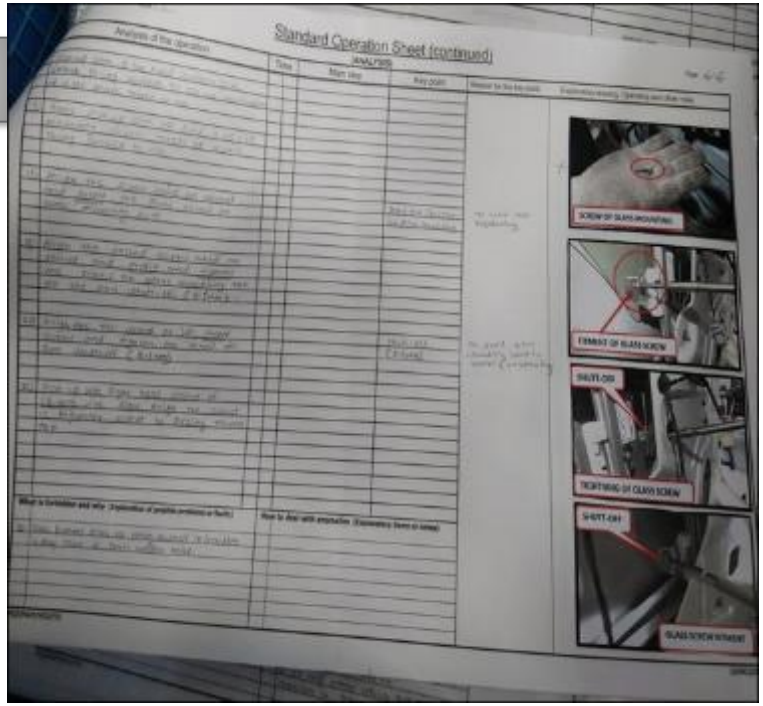
Rev No.:	Rev Date	Revision Summary	Reason for Revision
1	11-08-2016	Revision for VP-0	VP-0 Phase
2	18-07-2017	Revision for VP extension	VP Extension Phase
3	15.06.2018	Failure mode updated	Bi monthly review
4	09.09.2018	UPDATED FOR KHAT NOISE COMING FROM CO DRIVER POWER WINDOW DURING FULL UP TO DOWN-glass screw loose field concern	Field concern Sept-18

8. Standardization

8.1 Updation of documents :-

S N	Document	Target date	Responsibility	Status
5	SOS Updation (Rev No - 02)	25.08.18	Prasad Deore	CLOSED





8. Standardization

8.1 Updation of documents :-

S N	Document	Target date	Responsibility	Status
6	QA matrix updated	06.08.2018	Prasad Deore	Closed

QA MATRIX

PU NAME	Scoprio/Xylo
DEPT NAME	TCF
SECTION NAME	Trim 1

INITIALIZATION DATE	5/30/2016			
SIGNATURES				
Dept Head	Rahul Narkar			
Module Manager	Sandeep Chikale			
Section Manager	1	2	3	4

Global Quality Guarantee level.	
Section	0
Plant	100
Revision date	06.08.18

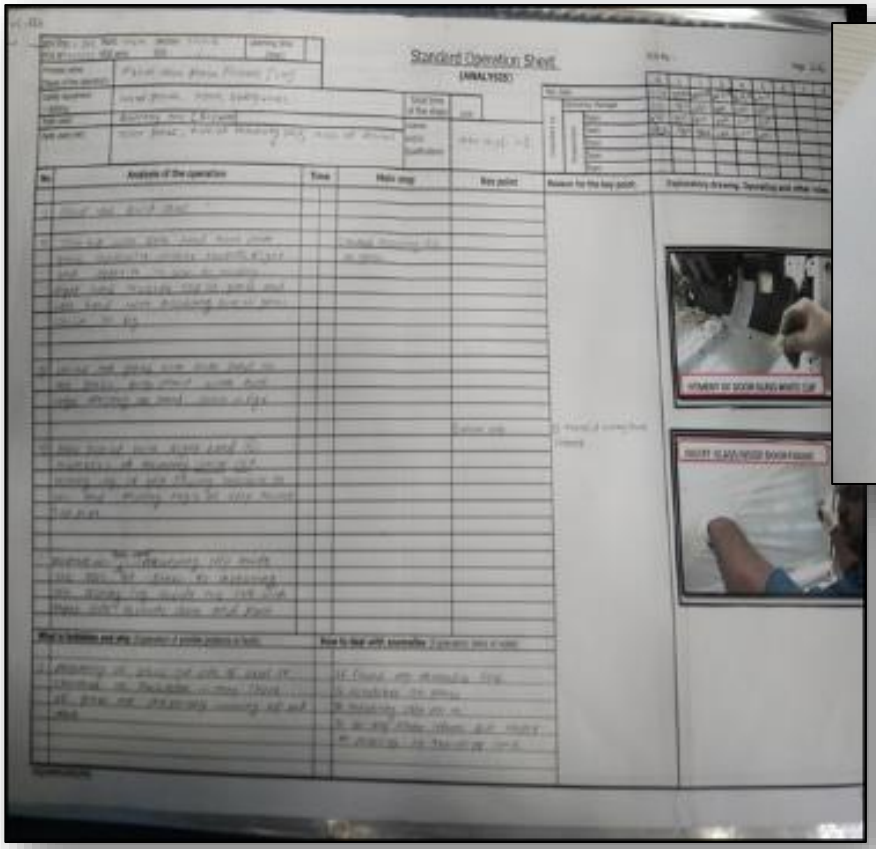
Workstation No.	Process description (POS)	Date	Failure modes	Severity	Prev. occurence	Potential deefect	Inspection process: Section																Inspection process: Outside section												Guarantee value				Faults recorded			Quality guarantee level				Comments
							Underbody	Regulator	Glass	IP	Airbag	BWT 3 buyoff											BWT 4 buyoff	BWT 6 buyoff	Roll down	Roller	tester	Shower	Model verification	S&R	Smoke	ECOS	CV	WRAP	ML		Workstation	Section	Outside section	PLANT	Section	Outside section	CUSTOMER	WORKSTATION	Section	Outside section	PLANT	
																																								1M	1M	3M					
	Glass fitment	06.08.18	Up/down noise	V2 3		●						⊙ 3																						□ 1			0	3	0	3				NOK	Ok	Ok	OK	

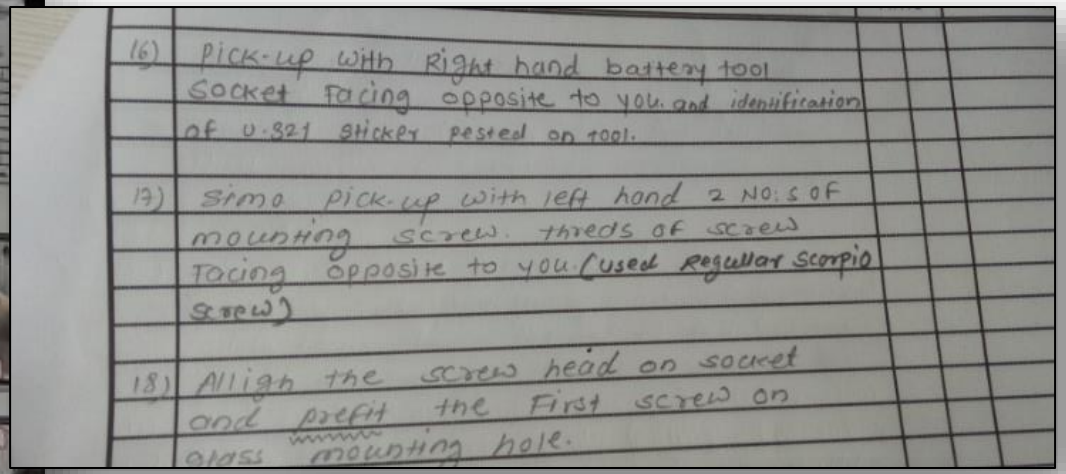
Note:-QA matrix at Workstation is red as No electric supply available on stage to check glass up down, This is ensured at Buy off.

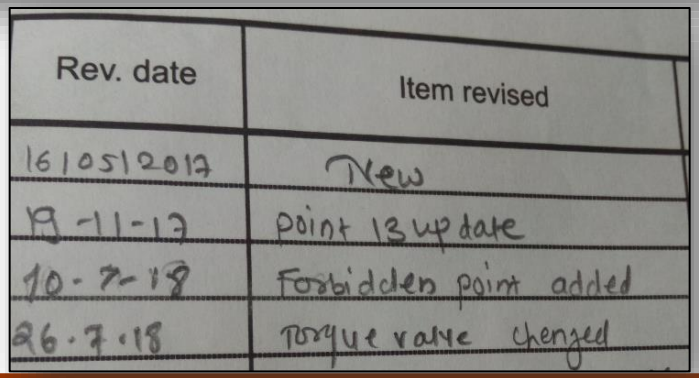
8. Standardization

8.2 Trauining-

SN	Training Identified	Target date	Responsibility	Status
1.	Training given to stage associates with updated SOS.	25.08.18	Prasad Deore	Closed







Rev. date	Item revised
16/05/2017	New
19-11-17	point 13 update
10-7-18	Forbidden point added
26.7.18	Torque value chenged

8. Standardization

8.3 Additional Audit & check

SN	Document	Target date	Responsibility	Status
1.	Process Audit check sheet	10.09.18	Satish Pagar	CLOSED

Marazzo - Process Audit check sheet (Trim1 Line)

Sr No	BWT	Stage / Fitment process	What to check
25	3	Water shield fitment	1. Water shield on triangular patch fitment as per req. position .
26	3	Glass seal fitment	corner matching during glass seal fitment
27	3	Glass fitment	correct tool used for glass tighteing (2.1 Nm)
28	3	Cluster flashing stage	1. All parts are kept in kitting bin (Not on floor) 2. Flashing SOS followed for cluster flashing
29	3	Triangular patch fitment	1. patch fitment by reference marking (Ensure no sliding of patch during fitment- Water shield may get torn)
	3		2. correct tightening sequence followed
30	3	IP fitment	1. Ip tightening sequence followed
			2. Use of fixture for IP insertion
	3		3. IP RH side wiring routing not on HVAC side . Wiring pushed away from HVAC on CCB
31	3	IP chiller hose fitment	1. No twisting of chiller hose during connection at IP stage 2.Check marking matching at both ends.
32	3	Shifter cable	1. Routine for shifter cable 2. Locking of shifter cable in its locking tower
33	3	Hood switch assly.	Hood switch rotating while locking
SPU/TCF/MQA/F/09			

9. Summarize & Plan future actions

9.1 Conclusion:

With provision of new screw & change in torque value , concern of all door glass up down noise eliminated..

9.2 Horizontal Deployment:

Inputs given to S201 project.

9.3 Lessons Learnt:

Things Gone Wrong : FMEA not captured interference of glass and clip must be more than 1 mm to avoid clearance after assembly.

Things Gone Right : Understood the importance of drawing study & GEMBA visit ,its mandatory consideration in process.

Thank you

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